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CONTINUOUS RANDOM VARIABLE

A continuous random variable is a variable that can assume an infinite, uncountable number of possible values within a given range or interval, typically representing measurements like time, weight, or distance.

PROBABILITY DENSITY FUNCTION

The probability density function of a continuous random variable can be defined as a function that gives the probability that the value of the random variable will fall between a range of values.

A continuous random variable X is described by a probability density function $f(x)$. The PDF $f(x)$ gives the relative likelihood of X taking on a specific value x.

The PDF must satisfy two conditions:

1. $f(x) \geq 0 \; for \; all \; x$
2. $\int_{-\infty}^{\infty} f(x)dx = 1.$

Cumulative Distribution Function (c.d.f.)

The cumulative distribution function $F(x)$ of a continuous random variable X represents the probability that X takes a value less than or equal to x:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

The CDF is non-decreasing and continuous, with: $\lim_{x \to -\infty} F(x) = 0 \; and \lim_{x \to \infty} F(x) = 1.$

MEAN OF CONTINUOUS RANDOM VARIABLE

The mean of a continuous random variable can be defined as the weighted average value of the random variable, X. It is also known as the expectation of the continuous random variable. The formula is given as follows

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx.$$

VARIANCE OF CONTINUOUS RANDOM VARIABLE

The variance of a continuous random variable can be defined as the expectation of the squared differences from the mean.

It helps to determine the dispersion in the distribution of the continuous random variable with respect to the mean. The formula is given as follows: $v(x) = E(x^2) - \big(E(x)\big)^2$ or

$$v(x) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx.$$

Example

1. A continuous random variable X has the following density function, $f(x) = kx(2-x)$ for $0 < x < 2$, find the value of (i) k (ii) Mean of X (iii) Variance of X.

Solution: Consider $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} 0dx + \int_{0}^{2} kx(2-x)dx + \int_{2}^{\infty} 0dx$$

$$1 = \int_0^2 (2kx - kx^2)dx$$

$$1 = k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left[4 - \frac{8}{3}\right] = k\frac{4}{3}$$

$$(i)\ k = \frac{3}{4}.$$

$$(ii)\ Mean\ E(x) = \int_{-\infty}^{\infty} xf(x)dx$$

$$E(x) = \int_{-\infty}^{0} 0\ dx + \int_0^2 x\frac{3}{4}x(2-x)\ dx + \int_2^{\infty} 0\ dx$$

$$E(x) = \frac{3}{4}\int_0^2 (2x^2 - x^3)\ dx$$

$$E(x) = \frac{3}{4}\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2$$

$$E(x) = \frac{3}{4}\left[\frac{16}{3} - 4\right] = 1.$$

$(ii)\ E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)dx$

$E(x^2) = \int_{-\infty}^{0} 0\ dx + \int_{0}^{2} x^2 \frac{3}{4}x(2-x)\ dx + \int_{2}^{\infty} 0\ dx$

$E(x^2) = \frac{3}{4}\int_{0}^{2} (2x^3 - x^4)\ dx$

$E(x^2) = \frac{3}{4}\left[\frac{2x^4}{4} - \frac{x^5}{5}\right]_0^2$

$E(x^2) = \frac{3}{4}\left[8 - \frac{32}{5}\right] = \frac{3}{4}.\frac{8}{5} = \frac{6}{5}$

$Variance = E(x^2) - \left(E(x)\right)^2 = \frac{6}{5} - 1 = \frac{1}{5}.$

2. A continuous random variable X has the following density function, $f(x) = cx^2$ for $|x| < 1$, find the value of (i) c (ii) Mean of X (iii) Variance of X (iv) $P\left(X \geq \frac{1}{2}\right)$.

Solution: Consider $\int_{-\infty}^{\infty} f(x)dx = 1$ and $|x| < 1\ i.e. -1 < x < 1$

$(i)\ Value\ of\ c$: $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{-1} 0dx + \int_{-1}^{1} cx^2 dx + \int_{1}^{\infty} 0dx$

$\therefore 1 = c\left[\frac{x^3}{3}\right]_{-1}^{1} = c\left[\frac{1}{3} + \frac{1}{3}\right] = c\frac{2}{3}$

$\therefore c = \frac{3}{2}.$

$(ii)\ Mean\ E(x) = \int_{-\infty}^{\infty} xf(x)dx$

$Mean\ E(x) = \int_{-\infty}^{-1} 0\ dx + \int_{-1}^{1} xcx^2\ dx + \int_{1}^{\infty} 0\ dx$

$E(x) = \frac{3}{2}\left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{3}{2}\left[\frac{1}{4} - \frac{1}{4}\right] = 0.$

$(iii)\ E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)dx$

$$E(x^2) = \int_{-\infty}^{-1} 0\,dx + \int_{-1}^{1} x^2 c x^2\,dx + \int_{1}^{\infty} 0\,dx$$

$$E(x^2) = \frac{3}{2}\int_{-1}^{1} (x^4)\,dx$$

$$E(x^2) = \frac{3}{4}\left[\frac{x^5}{5}\right]_{-1}^{1} = \frac{3}{2}\left[\frac{1}{5} + \frac{1}{5}\right] = \frac{3}{2}.\frac{2}{5} = \frac{3}{5}$$

$$v(x) = Variance = E(x^2) - \big(E(x)\big)^2 = \frac{3}{5} - 0 = \frac{3}{5}.$$

$$(iv)\ P\left(X \geq \frac{1}{2}\right) = \int_{-\infty}^{\infty} f(x)dx = \int_{\frac{1}{2}}^{1} cx^2 dx = \frac{3}{2}\left[\frac{x^3}{3}\right]_{\frac{1}{2}}^{1} = \frac{3}{2}\left[\frac{1}{3} - \frac{1}{24}\right] = \frac{7}{16}.$$

3. Find the expected value of a continuous random variable that follows a pdf given by

$$f(x) = \begin{cases} 3x & 0 \leq x \leq 1 \\ x+1 & 1 < x \leq 2 \\ 0 & otherwise \end{cases}$$

Solution: The formula for the expectation (Mean) of continuous random variable is

$$E(x) = \int_{-\infty}^{\infty} x f(x)dx$$

$$E(x) = \int_{-\infty}^{0} 0\, f(x)dx + \int_{0}^{1} x\, 3x\,dx + \int_{1}^{2} x\,(x+1)\,dx + \int_{2}^{\infty} 0\, f(x)dx$$

$$E(x) = 3\left[\frac{x^3}{3}\right]_0^1 + \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_1^2$$

$$E(x) = 1 - 0 + \frac{8}{3} + 2 - \frac{1}{3} - \frac{1}{2} = \frac{7}{3} + \frac{5}{2} = \frac{29}{6} = 4.833.$$

COMMON CONTINUOUS RANDOM VARIABLES

Some of the common distributions where continuous random variable is used are:

- Uniform Distribution
- Normal Distribution
- Exponential Distribution

UNIFORM (OR RECTANGULAR) DISTRIBUTION

A continuous random variable X is said to have a continuous rectangular (uniform) distributed over an interval (a, b), if its probability density function is given by:

$$f(x) = f(x; a, b) = \begin{cases} k & a < x < b \\ 0 & otherwise \end{cases} \qquad \ldots (*)$$

Given that $f(x)$ is *p.d.f.* Hence $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\Rightarrow \int_a^b k\, dx = 1 \Rightarrow k(b-a) = 1 \Rightarrow k = \frac{1}{b-a}.$$

Then the *p.d.f.* of X is given by $f(x) = f(x; a, b) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & otherwise \end{cases}$

Properties

1. $f(x) \geq 0$ for all x.
2. $\int_{-\infty}^{\infty} f(x)dx = 1$.

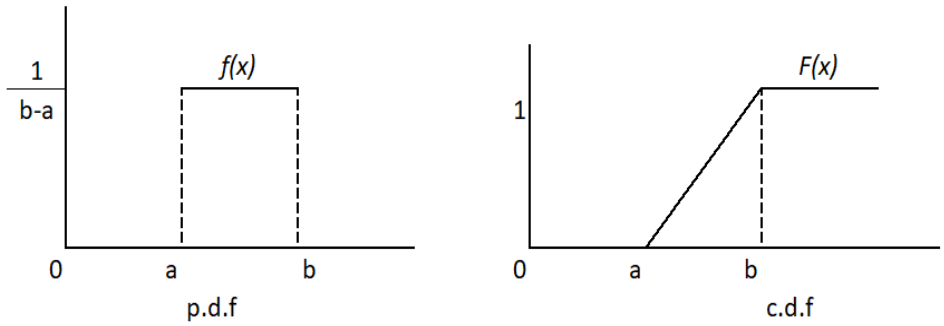
   $Proof: \int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} 0dx + \int_a^b \frac{1}{b-a}dx + \int_b^{\infty} 0dx = \frac{1}{b-a}[x]_a^b = \frac{b-a}{b-a} = 1.$
3. The distribution is also called as rectangular distribution, since the curve $y = f(x)$ describes a rectangle over the x-axis and between the ordinates $x = a$ and $x = b$.
4. If a and b, $(a < b)$ two parameters of the distribution. The distribution is called uniform distribution on (a, b) since it assumes a constant (uniform) value for all x in (a, b).
5. A uniform variate X on the interval (a, b) is written as $X \sim U[a, b]$ or $X \sim R[a, b]$.
6. The cumulative distribution function(c.d.f.) $F(x)$ is given by: $F(x) = \begin{cases} 0 & ; \quad x \le a \\ \frac{x-a}{b-a} & ; \quad a < x < b \\ 1 & ; \quad x \geq b \end{cases}$

   Since F(x) is not continuous and differentiable at $x = a$ and $x = b$.

   Thus $\frac{d}{dx}F(x) = \frac{d}{dx} = \frac{1}{b-a} \neq 0$ exists everywhere except at the points $x = a$ and $x = b$, and consequently probability density function $f(x)$ is given by equation $(*)$.
7. For a rectangular or uniform variate X in interval $(-a, a)$, the p.d.f. is given by

$$f(x) = f(x; -a, a) = \begin{cases} \frac{1}{2a} & ; -a < x < a \\ 0 & ; \ otherwise \end{cases} \qquad \ldots (*)$$

8. The graph of uniform p.d.f. and the corresponding cumulative distribution function $F(x)$ is



Mean of Uniform Distribution

Theorem: Let X be a random variable following a uniform distribution $X \sim U(a, b)$. Show that, the mean or expected value of X is $E(x) = \mu = \frac{a+b}{2}$.

Proof: Consider probability density function $f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & otherwise \end{cases}$

The expected value of uniform distribution X is $E(x) = \int_{-\infty}^{\infty} xf(x)dx$

$$E(x) = \int_{-\infty}^{0} xf(x)dx + \int_{a}^{b} xf(x)dx + \int_{b}^{\infty} xf(x)dx$$

$$E(x) = \int_{-\infty}^{0} 0\, dx + \int_{a}^{b} \frac{1}{b-a} x dx + \int_{b}^{\infty} 0\, dx$$

$$E(x) = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a}\left[\frac{b^2-a^2}{2}\right]$$

$$E(x) = \frac{1}{b-a}\left[\frac{(b-a)(b+a)}{2}\right]$$

$$E(x) = \frac{(a+b)}{2}.$$

Variance of Uniform Distribution

Theorem: Let X be a random variable following a uniform distribution $X \sim U(a, b)$ Then, show that, the variance of X is $v(x) = \sigma^2 = \frac{(b-a)^2}{12}$.

Proof: Consider probability density function $f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & otherwise \end{cases}$

The variance of uniform distribution X is $v(x) = E(x^2) - \big(E(x)\big)^2$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_{-\infty}^{0} x^2 f(x)dx + \int_{a}^{b} x^2 f(x)dx + \int_{b}^{\infty} x^2 f(x)dx$$

$$E(x^2) = \int_{-\infty}^{0} 0\ dx + \int_{a}^{b} \frac{1}{b-a} x^2 dx + \int_{b}^{\infty} 0\ dx$$

$$E(x^2) = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{1}{b-a}\left[\frac{b^3-a^3}{3}\right]$$

$$E(x^2) = \frac{1}{b-a}\left[\frac{(b-a)(b^2+ab+a^2)}{3}\right]$$

$$E(x^2) = \frac{(b^2+ab+a^2)}{3}$$

Variance $v(x) = E(x^2) - \big(E(x)\big)^2$

$$v(x) = \frac{(b^2+ab+a^2)}{3} - \left(\frac{a+b}{2}\right)^2$$

$$v(x) = \frac{b^2+ab+a^2}{3} - \frac{a^2+2ab+b^2}{4}$$

$$v(x) = \frac{4b^2+4ab+4a^2-3a^2-6ab-3b^2}{12}$$

$$v(x) = \frac{b^2-2ab+a^2}{12}$$

$$v(x) = \frac{(b-a)^2}{12} \ or\ \frac{(a-b)^2}{12}.$$

Examples

1. A bus arrives at a bus stop every 20 minutes. If you arrive at the bus stop at random time, what is the probability that you will have to wait more than 15 minutes for the bus?

Solution: Let X be the waiting time in minutes for bus. By using continuous uniform distribution with p.d.f. of X is $f(x) = \begin{cases} \frac{1}{20} & 0 < x < 20 \\ 0 & otherwise \end{cases}$ or $X \sim U[0,20]$

Use $P(x) = \int_{-\infty}^{\infty} f(x)dx$

The probability that, one will have to wait for more than 15 minutes is $P(x > 15)$

$$P(x > 15) = \int_{15}^{\infty} f(x)dx$$

$$P(x > 15) = \int_{15}^{20} f(x)dx + \int_{20}^{\infty} f(x)dx$$

$$P(x > 15) = \int_{15}^{20} \frac{1}{20}dx + \int_{20}^{\infty} 0\,dx$$

$$P(x > 15) = \frac{1}{20}[x]_{15}^{20}$$

$$P(x > 15) = \frac{1}{20}[20 - 15] = \frac{5}{20}$$

$$P(x > 15) = 0.25.$$

2. A machine produces parts that have a thickness uniformly distributed between 1.5 and 2.5 *mm*. what is the probability that a part has a thickness between 1.8 *mm* and 2.2 *mm*?

Solution: Let X be the part has a thickness between 1.8 mm and 2.2 mm. By using continuous uniform distribution with p.d.f. of X is $f(x) = \begin{cases} \frac{1}{2.5-1.5} = 1 & ; 1.5 < x < 2.5 \\ 0 & ; \quad otherwise \end{cases}$

i.e. $X \sim U[1.5,2.5]$

Use $P(x) = \int_{-\infty}^{\infty} f(x)dx$

The probability that a part thickness lies between 1.8 and 2.2 mm is

$$P(1.8 < x < 2.2) = \int_{-\infty}^{1.8} f(x)dx + \int_{1.8}^{2.2} f(x)dx + \int_{2.2}^{\infty} f(x)dx$$

$$P(1.8 < x < 2.2) = \int_{-\infty}^{1.8} 0\,dx + \int_{1.8}^{2.2} 1\,dx + \int_{2.2}^{\infty} dx$$

$$P(1.8 < x < 2.2) = [x]_{1.8}^{2.2} = 2.2 - 1.8$$

$$P(1.8 < x < 2.2) = 0.4.$$

3. Bus is uniformly late between 2 and 10 minutes. How long can you expect to wait? With what standard deviation? If there is a delay of more than 7 minutes, you will be late for work. What is the probability of you being late?

Solution: Let $a = 2, b = 10$

Mean of uniform distribution is $\mu = \frac{a+b}{2} = \frac{12}{2} = 6\ min$

Variance is $v(x) = \frac{(b-a)^2}{12} = \frac{64}{12} = 5.33\ min$ and Standard deviation is $\sigma = \sqrt{v} = 2.31 min.$

Probability density function is $f(x) = \begin{cases} \frac{1}{10-2} = \frac{1}{8} & ; 7 < x < 10 \\ 0 & ; \quad otherwise \end{cases}$

Required Probability is $P(7 \leq x \leq 10)$

$P(7 \leq x \leq 10) = \int_7^{10} \frac{1}{8} dx = \frac{1}{8}[10 - 7] = \frac{3}{8} = 0.375.$

4. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$, find $P(X < 0)$.

Solution: Given that $Mean = 1$ and $variance = \frac{4}{3}$

$Mean$: $1 = \frac{a+b}{2} \Rightarrow a + b = 2 \quad \ldots (i)$

$Variance$: $\frac{4}{3} = \frac{(b-a)^2}{12} \Rightarrow (b-a)^2 = 16 \Rightarrow b - a = \pm 4 \quad \ldots (ii)$

Adding eqn (i) and (ii), we get $2b = 2 \pm 4 \Rightarrow b = \frac{2 \pm 4}{2} \Rightarrow b = 3 \quad or\ b = -2$

If $b = -2, a = 4$ $(rejected\ because\ a \nless b)$

Therefore $b = 3$ and $a = -1$, and p.d.f is $f(x) = \begin{cases} \frac{1}{3-(-1)} = \frac{1}{4} & ; -1 < x < 3 \\ 0 & ; \quad otherwise \end{cases}$

$P(X < 0) = \int_{-1}^{0} f(x)dx = \int_{-1}^{0} \frac{1}{4} dx = \frac{1}{4}[0 + 1] = \frac{1}{4} = 0.25.$

5. Subway trains on a certain line run every half hour between mid-night and six in the morning. What is the probability that man entering the station at a random time during this period will have to wait at least 20 minutes?

Solution: Let X denote the waiting time for the next train. Under the assumption that a man arrives at the station at random, X is distributed uniformly on (0,30), with p.d.f.

$$f(x) = \begin{cases} \frac{1}{30 - 0} = \frac{1}{30} & ; 0 < x < 30 \\ 0 & ; \quad otherwise \end{cases}$$

The probability that, he has to wait at least 20 minutes is given by

$$P(x \geq 20) = \int_{20}^{\infty} f(x)dx = \int_{20}^{30} f(x)dx + \int_{30}^{\infty} f(x)dx = \int_{20}^{30} \frac{1}{30} dx + 0$$

$$P(x \geq 20) = \frac{1}{30}[x]_{20}^{30} = \frac{30 - 20}{30} = \frac{1}{3}.$$

6. Ace Heating and Air Conditioning Service finds that the amount of time a repairman needs to fix a furnace is uniformly distributed between 1.5 and four hours. Let the time needed to fix a furnace. Then.

a) Find the probability that a randomly selected furnace repair requires more than two hours.

b) Find the probability that a randomly selected furnace repair requires less than three hours.

c) Find the 30th percentile of furnace repair times.

d) The longest 25% of furnace repair times take at least how long? (In other words: find the minimum time for the longest 25% of repair times.) What percentile does this represent?

e) Find the mean and standard deviation.

Solution: Let $a = 1.5, b = 4$, uniform distribution with p.d.f. is

$$f(x) = \begin{cases} \frac{1}{4-(1.5)} = 0.4 & ; 1.5 < x < 4 \\ 0 & ; otherwise \end{cases}$$

a) Required Probability is: $P(x > 2)$

$$P(x > 2) = \int_2^{\infty} f(x)dx = \int_2^4 0.4dx + \int_4^{\infty} 0dx = 0.4[4 - 2] = 0.8.$$

b) Required Probability is: $P(x < 3)$

$$P(x < 3) = \int_{-\infty}^3 f(x)dx = \int_{-\infty}^{1.5} 0dx + \int_{1.5}^3 0.4dx = 0.4[3 - 1.5] = 0.6.$$

c) Required Probability is: $P(x < k)$

$$P(x < k) = \int_{-\infty}^k f(x)dx = \int_{-\infty}^{1.5} 0dx + \int_{1.5}^k 0.4dx = 0.4[k - 1.5]$$

$Put\ P(x < k) = 30\% = 0.30$

$\therefore 0.4[k - 1.5] = 0.30$

$\Rightarrow k = 2.25.$

The 30th percentile of repair times is 2.25 hours.

d) Required Probability is: $P(x > k)$

$$P(x > k) = \int_k^{\infty} f(x)dx = \int_k^4 0.4dx + \int_4^{\infty} 0dx = 0.4[4 - k]$$

$Given\ that, P(x > k) = 25\% = 0.25$

$\therefore 0.4[4 - k] = 0.25$

$\Rightarrow k = 3.375$

The longest 25% of furnace repairs take at least 3.375 hours (3.375 hours or longer).

e) Find the mean and standard deviation.

$Mean\ \mu = \frac{a+b}{2} = \frac{1.5+4}{2} = 2.75$

$Standard\ deviation\ is\ \sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(4-1.5)^2}{12}} 0.7217.$

EXPONENTIAL DISTRIBUTION

A continuous random variable X is said to have an exponential distribution with parameter (mean) $\mu > 0$, if its *p.d.f.* is: $f(x) = f(x,\mu) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ;\ for\ x \geq 0 \\ 0 & ;\ otherwise \end{cases}$

Properties

1. The *p.d.f.* is $f(x) \geq 0$ for all x, and the total area under the normal curve is unity. i.e. $\int_{-\infty}^{\infty} f(x)dx = 1$.

   $Proof: \int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} 0dx + \int_{0}^{\infty} \frac{1}{\mu} e^{-\frac{1}{\mu}x} dx = \frac{1}{\mu}\left[\frac{e^{-\frac{1}{\mu}x}}{-\frac{1}{\mu}}\right]_0^{\infty} = -e^{-\infty} + e^{-0} = 0 + 1 = 1.$

1. The cumulative distribution function $F(x)$ is given by:

   $F(x) = \int_0^x f(u)du = \int_0^x \frac{1}{\mu} e^{-\frac{1}{\mu}u} du = \frac{1}{\mu}\left[\frac{e^{-\frac{1}{\mu}u}}{-\frac{1}{\mu}}\right]_0^{x} = 1 - e^{-\frac{1}{\mu}x}$

   $F(x) = \begin{cases} 1 - e^{-\frac{1}{\mu}x} & ;\ for\ x \geq 0 \\ 0 & ;\ otherwise \end{cases}$

Mean of Exponential Distribution

Theorem: Let X be a random variable following an exponential distribution $X \sim exp(\mu)$. Show that, the mean or expected value of X is $E(x) = \mu$.

Proof: Consider probability density function $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ;\ for\ x \geq 0 \\ 0 & ;\ otherwise \end{cases}$

The expected value (Mean) of exponential distribution X is $E(x) = \int_{-\infty}^{\infty} xf(x)dx$

$E(x) = \int_{-\infty}^{0} xf(x)dx + \int_{0}^{\infty} xf(x)dx$

$$E(x) = \int_{-\infty}^{0} 0\, dx + \int_{0}^{\infty} x \frac{1}{\mu} e^{-\frac{1}{\mu}x}\, dx$$

$$E(x) = \frac{1}{\mu}\left[x \frac{e^{-\frac{1}{\mu}x}}{-\frac{1}{\mu}} - \frac{e^{-\theta x}}{\left(\frac{1}{\mu}\right)^2} \right]_0^{\infty} = \frac{1}{\mu}[0 + 0 + 0 + \mu^2]$$

$E(x) = \mu.$

Variance of Exponential Distribution

Theorem: Let X be a random variable following an exponential distribution $X \sim exp(\theta)$ Then, show that, the variance of X is $v(x) = \mu^2$.

Proof: Consider probability density function $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; otherwise \end{cases}$

The variance of exponential distribution X is $v(x) = E(x^2) - \left(E(x)\right)^2$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{0} x^2 f(x) dx + \int_{0}^{\infty} x^2 f(x) dx$$

$$E(x^2) = \int_{-\infty}^{0} 0\, dx + \int_{0}^{\infty} x^2 \frac{1}{\mu} e^{-\frac{1}{\mu}x}\, dx$$

$$E(x^2) = \frac{1}{\mu}\left[x^2 \frac{e^{-\frac{1}{\mu}x}}{-\frac{1}{\mu}} - 2x \frac{e^{-\frac{1}{\mu}x}}{\left(\frac{1}{\mu}\right)^2} + 2 \frac{e^{-\frac{1}{\mu}x}}{-\left(\frac{1}{\mu}\right)^3} \right]_0^{\infty}$$

$$E(x^2) = \frac{1}{\mu}[0 - 0 + 0 - 0 + 0 + 2\mu^3] = 2\mu^2$$

Variance $v(x) = E(x^2) - \left(E(x)\right)^2$

$v(x) = 2\mu^2 - \mu^2$

$v(x) = \mu^2$

Examples

1. The lifetime of a light bulb follows an exponential distribution with a mean of 1000 hrs. what is the probability that a light bulb will last more than 1500 hrs.?

   Solution: Let X be the lifespan of a light bulb in hours. Given that the mean of exponential distribution i.e. $\mu = 1000$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{1000} e^{-\frac{x}{1000}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

The probability that, light bulb will last more than 1500 hours is $P(x \geq 1500)$

$$P(x \geq 1500) = \int_{1500}^{\infty} f(x)dx$$

$$P(x \geq 1500) = \int_{1500}^{\infty} \frac{1}{1000} e^{-\frac{x}{1000}} dx$$

$$P(x \geq 1500) = \frac{1}{1000}\left[\frac{e^{-\frac{x}{1000}}}{-\frac{1}{1000}}\right]_{1500}^{\infty}$$

$$P(x \geq 1500) = \left[-e^{-\frac{x}{1000}}\right]_{1500}^{\infty} = [-e^{-\infty} + e^{-1.5}] = e^{-1.5} = 0.2231.$$

Therefore, the probability that, light bulb will last more than 1500 hours is 0.2231.

2. The time between failures of a machine follows an exponential distribution with a mean of 50 hrs. what is the probability that, the machine will fail within the next 20 hrs.?

Solution: Let X be the time between failures of a machine in hours. Given that the mean of exponential distribution i.e. $\mu = 50$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{50} e^{-\frac{x}{50}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

The probability that, the machine will fail within the next 20 hours is $P(0 \leq x \leq 20)$

$$P(0 \leq x \leq 20) = \int_{0}^{20} f(x)dx$$

$$P(0 \leq x \leq 20) = \int_{0}^{20} \frac{1}{50} e^{-\frac{x}{50}} dx$$

$$P(0 \le x \le 20) = \frac{1}{50}\left[\frac{e^{-\frac{x}{50}}}{-\frac{1}{50}}\right]_0^{20}$$

$$P(0 \le x \le 20) = \left[-e^{-\frac{x}{50}}\right]_0^{20} = \left[-e^{-\frac{20}{50}} + e^{0}\right] = 1 - 0.6703 = 0.3297.$$

Therefore, the probability that, the machine will fail within the next 20 hours is 0.3297.

3. The lifetime of a battery follows exponential distribution with mean of 200 hrs. what is the probability that, the battery will last for more than 300hrs.?

Solution: Let X be the lifespan of a battery in hours. Given that the mean of exponential distribution i.e. $\mu = 200$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \ge 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{200} e^{-\frac{x}{200}} & ; for\ x \ge 0 \\ 0 & ; x < 0 \end{cases}$$

The probability that, battery will last more than 300 hours is $P(x \ge 300)$

$$P(x \ge 300) = \int_{300}^{\infty} f(x)dx$$

$$P(x \ge 300) = \int_{300}^{\infty} \frac{1}{200} e^{-\frac{x}{200}} dx$$

$$P(x \ge 300) = \frac{1}{200}\left[\frac{e^{-\frac{x}{200}}}{-\frac{1}{200}}\right]_{300}^{\infty}$$

$$P(x \ge 300) = \left[-e^{-\frac{x}{200}}\right]_{300}^{\infty} = [-e^{-\infty} + e^{-1.5}] = e^{-1.5} = 0.2231.$$

Therefore, the probability that, battery will last more than 300 hours is 0.2231.

4. On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed. (a) What is the probability that a computer part lasts more than 7 years? (b) What is the probability that a computer part lasts between 9 and 11 years?

Solution: Let X be the lifespan of a computers parts in years. Given that the mean of exponential distribution i.e. $\mu = 10$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{10} e^{-\frac{x}{10}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

(a) The probability that, a computer part lasts more than 7 years is $P(x \geq 7)$

$$P(x \geq 7) = \int_7^{\infty} f(x)dx$$

$$P(x \geq 7) = \int_7^{\infty} \frac{1}{10} e^{-\frac{x}{10}} dx$$

$$P(x \geq 7) = \frac{1}{10}\left[\frac{e^{-\frac{x}{10}}}{-\frac{1}{10}}\right]_7^{\infty}$$

$$P(x \geq 7) = \left[-e^{-\frac{x}{10}}\right]_7^{\infty} = \left[-e^{-\infty} + e^{-\frac{7}{10}}\right] = e^{-0.7} = 0.4966.$$

Therefore, the probability that, a computer part lasts more than 7 years is 0.4966.

(b) The probability that, a computer part lasts more than 7 years is $P(9 < x < 11)$

$$P(9 < x < 11) = \int_9^{11} f(x)dx$$

$$P(9 < x < 11) = \int_9^{11} \frac{1}{10} e^{-\frac{x}{10}} dx$$

$$P(9 < x < 11) = \frac{1}{10}\left[\frac{e^{-\frac{x}{10}}}{-\frac{1}{10}}\right]_9^{11}$$

$$P(9 < x < 11) = \left[-e^{-\frac{11}{10}} + e^{-\frac{9}{10}}\right] = 0.0737.$$

Therefore, the probability that, a computer part lasts more than 7 years is 0.0737.

5. The number of days ahead travelers purchase their airline tickets can be modeled by an exponential distribution with the average amount of time equal to 15 days. Find the probability that a traveler will purchase a ticket fewer than ten days in advance.?

Solution: Let X be the travelers purchase their airline tickets in days. Given that the mean of exponential distribution i.e. $\mu = 15$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{15} e^{-\frac{x}{15}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

The probability that a traveler will purchase a ticket fewer than ten days in advance is $P(x < 10)$

$$P(x < 10) = \int_0^{10} f(x)dx$$

$$P(x < 10) = \int_0^{10} \frac{1}{15} e^{-\frac{x}{15}} dx$$

$$P(x < 10) = \frac{1}{15} \left[\frac{e^{-\frac{x}{15}}}{-\frac{1}{15}} \right]_0^{10}$$

$$P(x < 10) = \left[-e^{-\frac{10}{15}} + e^{-0} \right] = 1 - e^{-0.67} = 0.4866.$$

Therefore, the probability that a traveler will purchase a ticket fewer than ten days in advance is 0.4866.

6. Let X = amount of time (in minutes) a postal clerk spends with his or her customer. The time is known to have an exponential distribution with the average amount of time equal to four minutes. Find the probability that a clerk spends four to five minutes with a randomly selected customer?

Solution: Given that the mean of exponential distribution i.e. $\mu = 4$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{4} e^{-\frac{x}{4}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

The probability that a clerk spends four to five minutes with a randomly selected customer is $P(4 < x < 5)$:

$$P(4 < x < 5) = \int_4^5 f(x)dx$$

$$P(4 < x < 5) = \int_4^5 \frac{1}{4} e^{-\frac{x}{4}} dx$$

$$P(4 < x < 5) = \frac{1}{4}\left[\frac{e^{-\frac{x}{4}}}{-\frac{1}{4}}\right]_4^5$$

$$P(4 < x < 5) = \left[-e^{-\frac{5}{4}} + e^{-1}\right] = 1 - e^{-0.67} = 0.0814.$$

Therefore, the probability that a clerk spends four to five minutes with a randomly selected customer is 0.0814.

7. The mileage which car owners get with a certain kind of radial tyre is a random variable having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last (i) at least 20,000 km and (ii) at most 30,000 km.?

Solution: Given that the mean of exponential distribution i.e. $\mu = 40000$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{40000} e^{-\frac{x}{40000}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

(a) The probability that, one of these tires will last at least 20,000 km is $P(x \geq 20000)$

$$P(20000 \leq x) = \int_{20000}^{\infty} f(x)dx$$

$$P(20000 \leq x) = \int_{20000}^{\infty} \frac{1}{40000} e^{-\frac{x}{40000}} dx$$

$$P(20000 \leq x) = \frac{1}{40000}\left[\frac{e^{-\frac{x}{40000}}}{-\frac{1}{40000}}\right]_{20000}^{\infty}$$

$$P(20000 \leq x) = \left[-e^{-\infty} + e^{-\frac{1}{2}}\right] = 0 + e^{-0.5} = 0.6065.$$

Therefore, the probability that, one of these tires will last at least 20,000 km is 0.6065.

(b) The probability that, one of these tires will last at most 30,000 km is $P(x \leq 30000)$

$$P(x \leq 20000) = \int_0^{30000} f(x)dx$$

$$P(x \leq 20000) = \int_0^{30000} \frac{1}{40000} e^{-\frac{x}{40000}} dx$$

$$P(x \leq 20000) = \frac{1}{40000}\left[\frac{e^{-\frac{x}{40000}}}{-\frac{1}{40000}}\right]_0^{30000}$$

$$P(x \leq 20000) = \left[-e^{-\frac{3}{4}} + e^{-0}\right] = 1 - e^{-0.75} = 0.5276.$$

Therefore, the probability that, one of these tires will last at most 30,000 km is 0.5276.

8. The length of time a person speaks over phone follows exponential distribution with mean 6. What is the probability that the person will talk for (1) more than 8 minutes (2) between 4 and 8 minutes?

Solution: Given that the mean of exponential distribution i.e. $\mu = 6$.

The probability density function of X is $f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{1}{\mu}x} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$

$$f(x) = \begin{cases} \frac{1}{6} e^{-\frac{x}{6}} & ; for\ x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

1) The probability that, the person will talk for more than 8 minutes is $P(x \geq 8)$

$$P(x > 8) = \int_8^{\infty} f(x)dx$$

$$P(x > 8) = \int_8^{\infty} \frac{1}{6} e^{-\frac{x}{6}} dx$$

$$P(x > 8) = \frac{1}{6}\left[\frac{e^{-\frac{x}{6}}}{-\frac{1}{6}}\right]_8^{\infty}$$

$$P(x > 8) = \left[-e^{-\infty} + e^{-\frac{8}{6}}\right] = 0 + e^{-1.33} = 0.2636.$$

Therefore, the probability that, the person will talk for more than 8 minutes is 0.2636.

2) The probability that, the person will talk between 4 and 8 minutes is $P(4 < x < 8)$

$$P(4 < x < 8) = \int_4^8 f(x)dx$$

$$P(4 < x < 8) = \int_4^8 \frac{1}{6} e^{-\frac{x}{6}} dx$$

$$P(4 < x < 8) = \frac{1}{6}\left[\frac{e^{-\frac{x}{6}}}{-\frac{1}{6}}\right]_4^8$$

$$P(4 < x < 8) = \left[-e^{-\frac{8}{6}} + e^{-\frac{4}{6}}\right] = 0.2499.$$

Therefore, the probability that, the person will talk between 4 and 8 minutes is 0.2499.

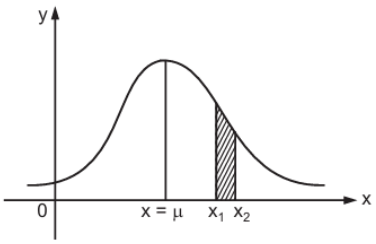
NORMAL DISTRIBUTION

Normal distribution is obtained as a limiting form of Binomial distribution when n the number of trials is very large and neither p nor q is very small. Most of the modern statistical methods have been based on this distribution.

Definition: A continuous random variable X is said to follow a normal distribution with mean μ and variance σ^2, if its *p.d.f.* is given by,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2};\ -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0.$$

The probability that the continuous random variable x lies between $x = x_1$ and $x = x_2$ is given by the area under the curve $y = f(x)$ bounded by x-axis, $x = x_1$ and $x = x_2$ which is shown by shaded area in the Fig.

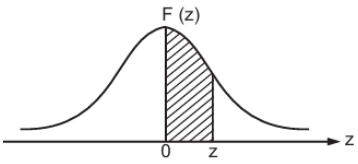


$$P(x_1 \le x \le x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx.$$

Introducing $z = \frac{x-\mu}{\sigma}$, the above integral becomes independent (dimensionless) of two parameters μ and σ. Here z is known as standard variable (variate). With $z = \frac{x-\mu}{\sigma}$, $dx = \sigma\, dz$ and $z_1 = \frac{x_1-\mu}{\sigma}$, $z_2 = \frac{x_2-\mu}{\sigma}$, hence above integral is

$$P(z_1 \le z \le z_2) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}}\, dz.$$

If $\mu = 0$ and $\sigma = 1$ then standard variable $z = \frac{x-\mu}{\sigma}$ is called standard normal variable i.e. $N(0,1)$. The p.d.f. of standard normal random variable is given by $F(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$.



If $z = \frac{x-\mu}{\sigma} \to N(0,1)$ then the distribution is called **standard normal distribution.** and its normal curve as standard normal cure (refer above fig.).

The probability of random variable x lying between $x = \mu$ and any value of $x = x_1$ is given by;

$$P(\mu \le x \le x_1) = \int_{\mu}^{x_1} f(x)dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^{x_1} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}\, dz$$

When $z = \frac{x-\mu}{\sigma}$, $dx = \sigma\, dz$, and $x = \mu, z = 0$, and $x = x_1, z = \frac{x_1-\mu}{\sigma} = z_1(say)$

$$P(0 \le z \le z_1) = \frac{1}{\sqrt{2\pi}} \int_{0}^{z_1} e^{-\frac{z^2}{2}}\, dz.$$

If A(z) denotes the area (refer above fig.) under the normal curve $F(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$. from 0 to z_1, z_1 being any number then $P(z) = A(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-\frac{z^2}{2}}\, dz$.

Result; $P(x_1 \le x \le x_2) = P(z_1 \le z \le z_2) = P(z_2) - P(z_1)$

$P = (Area\ under\ the\ normal\ curve\ from\ 0\ to\ z_2) - (Area\ under\ the\ normal\ curve\ from\ 0\ to\ z_1)$

Properties

2. In normal distribution Mean, Median, and Mode all are equal.
3. The graph of $f(x)$ is *'bell-shaped'* curve.
4. When a random variable is normally distributed with mean μ and standard deviation σ, it is customary to write X is distributed as $N(\mu, \sigma^2)$ and is expressed by $X \sim N(\mu, \sigma^2)$.
5. The *p.d.f.* of standard normal variate z is $\varphi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{z^2}{2}};\ -\infty < z < \infty,$ the corresponding distribution function is denoted by $\Phi(z)$ and it is given by $\Phi(z) = P(Z \le z)$

   $\Phi(z) = \int_{-\infty}^{z} \varphi(u)du = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{u^2}{2}} du = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{u^2}{2}} du$ and $\Phi(-z) = 1 - \Phi(z);\ z > 0.$

6. The *p.d.f.* is $f(x) \geq 0$ for all x, and the total area under the normal curve is unity. i.e. $\int_{-\infty}^{\infty} f(x)dx = 1$.

Proof: $\int_{-\infty}^{\infty} f(x)dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \frac{dx}{\sigma}$ $Put\ z = \frac{x-\mu}{\sigma} \Rightarrow dz = \frac{1}{\sigma}dx$

$\int_{-\infty}^{\infty} f(x)dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz$, ... But $e^{-\frac{z^2}{2}}$ is even function,

$$\int_{-\infty}^{\infty} f(x)dx = \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{z^2}{2}} dz$$

By using result $\int_0^{\infty} z^{n-1} e^{-az^2} dz = \frac{\Gamma\left(\frac{n}{2}\right)}{2\, a^{\frac{n}{2}}}$

$$\int_{-\infty}^{\infty} f(x)dx = \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} z^{1-1} e^{-\frac{1}{2}z^2} dz = \frac{2}{\sqrt{2\pi}} \frac{\Gamma\left(\frac{1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{1}{2}}}$$

$$\int_{-\infty}^{\infty} f(x)dx = \frac{1}{\sqrt{2\pi}} \frac{[\sqrt{\pi}]}{1} (2)^{\frac{1}{2}}$$

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

Mean of Normal Distribution

Theorem: Let X be a random variable following a normal distribution $X \sim N(\mu, \sigma^2)$. Show that, the mean or expected value of X is $E(x) = \mu$.

Proof: The *p.d.f.* of normal distribution is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ and the expected value (Mean) of normal distribution X is

$E(x) = \int_{-\infty}^{\infty} x f(x)dx$

$$E(x) = \int_{-\infty}^{\infty} x\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \frac{dx}{\sigma}$$

$Put\ z = \frac{x-\mu}{\sigma} \Rightarrow dz = \frac{1}{\sigma}dx$ and $x - \mu = z\sigma \Rightarrow x = \mu + z\sigma$

$$E(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + z\sigma)\, e^{-\frac{z^2}{2}} dz = \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z\, e^{-\frac{z^2}{2}} dz$$

$Here, z\,e^{-\frac{z^2}{2}}$ *is odd function and* $e^{-\frac{z^2}{2}}$ *is even function*

$$E(x) = \frac{2\mu}{\sqrt{2\pi}}\int_0^{\infty} e^{-\frac{z^2}{2}}dz + 0$$

By using result $\int_0^{\infty} z^{n-1}\,e^{-az^2}dz = \frac{\Gamma\left(\frac{n}{2}\right)}{2\,a^{\frac{n}{2}}} \rightarrow \int_0^{\infty} z\,e^{-\frac{1}{2}z^2}dz = \frac{\Gamma\left(\frac{1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{1}{2}}} = \frac{\sqrt{\pi}}{2\sqrt{\frac{1}{2}}} = \frac{\sqrt{\pi}}{\sqrt{2}}$

$$E(x) = \frac{2\mu}{\sqrt{2\pi}}\int_0^{\infty} z^{1-1}\,e^{-\frac{1}{2}z^2}dz = 2\frac{\mu}{\sqrt{2\pi}}\frac{\sqrt{\pi}}{\sqrt{2}}$$

$$E(x) = \mu.$$

VARIANCE OF NORMAL DISTRIBUTION

Theorem: Let X be a random variable following a normal distribution $X \sim N(\mu, \sigma^2)$. Show that, the variance of X is $v(x) = \sigma^2$.

Proof: The *p.d.f.* of normal distribution is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ and the variance of normal distribution X is $v(x) = E[X - E(x)]^2 = E(x-\mu)^2$

$$v(x) = \int_{-\infty}^{\infty}(x-\mu)^2 f(x)dx$$

$$v(x) = \int_{-\infty}^{\infty}(x-\mu)^2\frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

$Put\ z = \frac{x-\mu}{\sigma} \Rightarrow dz = \frac{1}{\sigma}dx\ and\ x-\mu = z\sigma$

$$v(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\sigma^2 z^2\,e^{-\frac{z^2}{2}}dz = \frac{\sigma^2}{\sqrt{2\pi}}\int_{-\infty}^{\infty} z^2\,e^{-\frac{z^2}{2}}dz \quad \ldots but\ z^2 e^{-\frac{z^2}{2}}\ is\ even\ function$$

$$v(x) = 2\frac{\sigma^2}{\sqrt{2\pi}}\int_0^{\infty} z^2\,e^{-\frac{z^2}{2}}dz$$

By using result $\int_0^{\infty} z^{n-1}\,e^{-az^2}dz = \frac{\Gamma\left(\frac{n}{2}\right)}{2\,a^{\frac{n}{2}}} \rightarrow \int_0^{\infty} z^2\,e^{-\frac{z^2}{2}}dz = \frac{\Gamma\left(\frac{3}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{3}{2}}} = \frac{\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{\sqrt{\frac{1}{2}}} = \frac{\sqrt{\pi}}{\sqrt{2}}$

$$v(x) = 2\frac{\sigma^2}{\sqrt{2\pi}}\frac{\sqrt{\pi}}{\sqrt{2}}$$

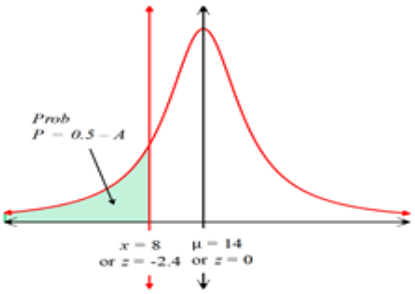
$v(x) = \sigma^2$.

Examples

1. In a sample of 1000 candidates, the mean of certain test is 14 and standard deviation is 2.5. Assuming Normal distribution, find the number of candidates getting less than eight marks. [Given that z=2.4, A = 0.4918].

Solution: Let Mean $\mu = 14$, Standard deviation $\sigma = 2.5$, and $x = 8$ $So, z = \frac{x-\mu}{\sigma} = -2.4$

The probability that candidates getting less than eight marks= $P(X < 8)$



$= P\left(\frac{x-14}{2.5} < \frac{8-14}{2.5}\right)$

$= P(z < -2.4) = P(z \geq 2.4)$

$= 0.5 - P(0 \leq z \leq 2.4)$

$= 0.5 - 0.4918 = 0.0082$

So, the expected number of candidates getting less than 8 marks out of 1,000 candidates is

$= 1000\ P(X < 8)$

$= 1000 * 0.0082$

$= 8.2 \approx 8.$

2. In a normally distributed group of 450 students with mean 42 and standard deviation 8, find the number of students scoring less than 48 marks. [Given that z= 0.75, A = 0.2734].

Solution: Let Mean $\mu = 42$, Standard deviation $\sigma = 8$, and $x = 48$

$So, z = \frac{x-\mu}{\sigma} = \frac{48-42}{8} = 0.75$

The probability of students scoring less than 48 marks, $= P(X < 48)$

$= P\left(\frac{x-42}{8} < \frac{48-42}{8}\right)$

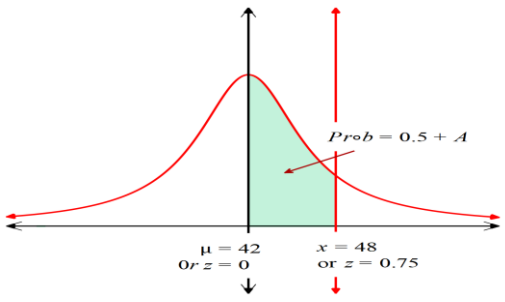
$= P(z < 0.75) = P(-\infty \leq z \leq 0 \leq 0.75)$

$= P(-\infty \leq z \leq 0) + P(0 \leq z \leq 0.75)$

$= 0.5 + P(0 \leq z \leq 0.75)$

$= 0.5 + 0.2734 = 0.7734$

So, the expected number of students scoring less than 48 marks out of 450 students



$= 450\ P(X < 48)$

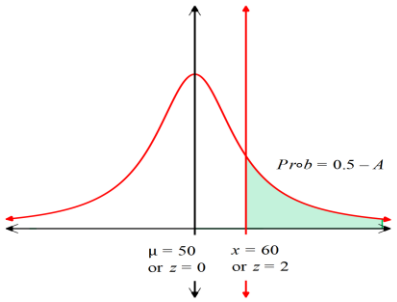
$= 450 \times 0.7734 = 348.03 \approx 348.$

3. In a certain examination test 10000 students appeared in a paper of mathematics. Average marks obtained were 50% with Standard Deviation 5%. Marks are normally distributed. Find the number of students expected to get more than 60% marks. [Given that z=2, A=0.4772].

Solution: Let Mean $\mu = 50\%$, Standard deviation $\sigma = 5\%$, and $x = 60\%$

$So, z = \frac{x-\mu}{\sigma} = \frac{60-50}{5} = 2$

The probability of students expected to get more than 60% marks= $P(X > 60)$



$= P\left(\frac{x-50}{5} > \frac{60-50}{5}\right)$

$= P(z > 2) = 0.5 - P(0 \le z \le 2)$

$= 0.5 - 0.4772 = 0.0228$

So, the expected number of students to get more than 60% marks out of 10000 students.

$= 10000\ P(X > 60)$

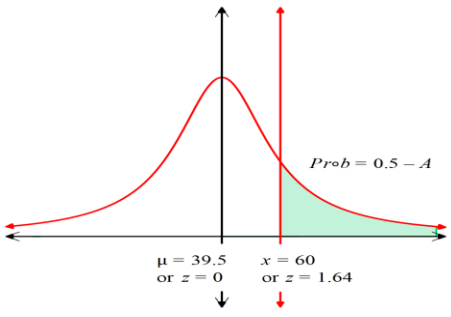
$= 10000 \times 0.0228 = 228.$

4. 5000 candidates appeared in a certain paper carrying a maximum of 100 marks. It was found that marks were normally distributed with mean 39.5 and standard deviation 12.5. Determine the number of candidates who secured a first class for which a minimum of 60 marks is necessary. [Given that z= 1.64, A=0.4495].

Solution: Let Mean $\mu = 39.5$, Standard deviation $\sigma = 12.5$, and $x = 60$

$So, z = \frac{x-\mu}{\sigma} = \frac{60-39.5}{12.5} = 2$

The probability of candidates who secured a first class for which a minimum of 60 marks is necessary= $P(X > 60)$



$= P\left(\frac{x-39.5}{12.5} > \frac{60-39.5}{12.5}\right)$

$= P(z > 1.64) = 0.5 - P(0 \le z \le 1.64)$

$= 0.5 - 0.4495 = 0.0505$

So, the expected number of candidates who secured a first class for which a minimum of 60 marks are necessary= $5000\ P(X > 60)$

$= 5000 \times 0.0505$

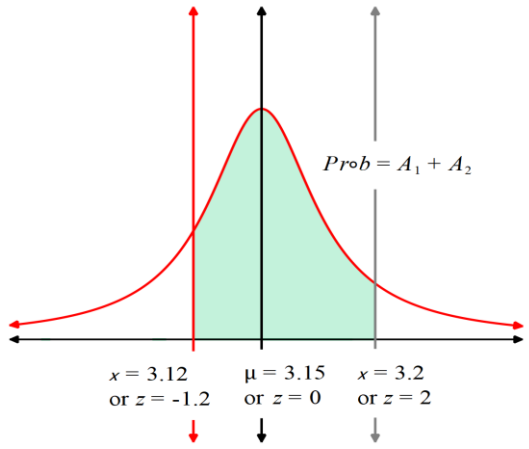
$= 252.5 \approx 253.$

5. A random sample of 200 screws is drawn from a population which represents the size of screws. If a sample is distributed normally with mean 3.15 cm and standard deviation 0.025 cm, find expected number of screws whose size falls between 3.12 cm and 3.2 cm. [Given that z=1.2, A=0.3849 and z=2, A=0.4772].

Solution: Let Mean $\mu = 3.15$, Standard deviation $\sigma = 0.025$, and $Let\ x_1 = 3.12,\ x_2 = 3.2$

$So, z_1 = \frac{x_1-\mu}{\sigma} = \frac{3.12-3.15}{0.025} = -1.2 \quad and$

$z_2 = \frac{x_2-\mu}{\sigma} = \frac{3.2-3.15}{0.025} = 2$



The probability of screws whose size falls between 3.12 and 3.2 cm is

$= P(3.12 < X < 3.2)$

$= P\left(\frac{3.12-3.15}{0.025} < \frac{x-3.15}{0.025} < \frac{3.2-3.15}{0.025}\right)$

$= P(-1.2 < z < 2)$

$= P(-1.2 \le z \le 0) + P(0 \le z \le 2)$

$= P(0 \le z \le 1.2) + P(0 \le z \le 2)$

$= 0.3849 + 0.4772 = 0.8621$

So, the expected number of screws whose size falls between 3.12 cm and 3.2 cm is

$= 200\ P(3.12 < X < 3.2) = 200 \times 0.8621$

$= 172.42 \approx 172.$

6. The mean weight of 500 students is 63kg and standard deviation is 8kg. Assuming that the weights are normally distributed, find how many students weigh 52kg? The weights are recorded to the nearest kg.

[Given that, $z = 1.44, A = 0.4251$ and for $z = 1.31, A = 0.4049$].

Solution: Let Mean $\mu = 63$, Standard deviation $\sigma = 8$, and $Let\ x_1 = 51.5\ \ and\ x_2 = 52.5$ (nearest 5kg weight)

$z_1 = \frac{x_1-\mu}{\sigma} = \frac{51.5-63}{8} = -1.44\ \ and\ z_2 = \frac{x_2-\mu}{\sigma} = \frac{52.5-63}{8} = -1.31$

Required probability is $= P(51.5 < X < 52.5)$

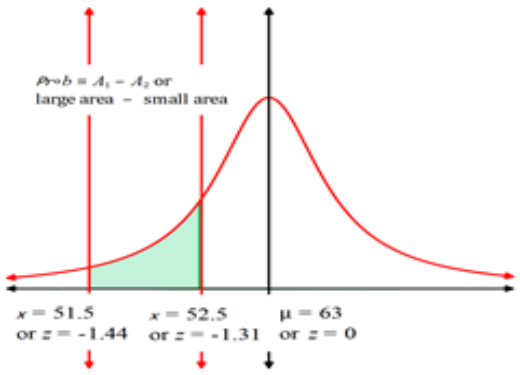
$= P\left(\frac{51.5-63}{8} < \frac{x-63}{8} < \frac{52.5-63}{8}\right)$

$= P(-1.44 < z < -1.31)$

$= P(1.31 \leq z \leq 1.44)$

$= P(0 \leq z \leq 1.44) - P(0 \leq z \leq 1.31)$

$= 0.4251 - 0.4049 = 0.0202$



So, the expected number of students weigh 52kg

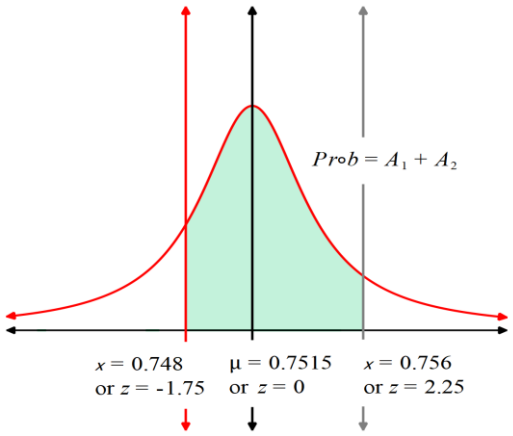
$= 500\ P(51.5 < X < 52.5)$

$= 500 \times 0.0202 = 10.1 \approx 10.$

7. Assuming that, the diameter of 1000 brass plugs taken consecutively from machine form a normal distribution with mean 0.7515 cm and standard deviation 0.002 cm. How many of the plugs are likely to be appeared if the acceptable diameter is 0.752 ± 0.004.

[Given that, z= 2.25, A= 0.4878 and z=1.75, A=0.4599].

Solution: Let mean $\mu = 0.7515$, and standard deviation $\sigma = 0.002$ $Let\ x_1 = 0.752 + 0.004 = 0.756\ and\ x_2 = 0.752 - 0.004 = 0.748$



$z_1 = \frac{x_1-\mu}{\sigma} = \frac{0.756-0.7515}{0.002} = 2.25\ and$

$z_2 = \frac{x_2-\mu}{\sigma} = \frac{0.748-0.7515}{0.002} = -1.75$

Required probability is

$P = P(0.748 < X < 0.756)$

$= P\left(\frac{0.748-0.7515}{0.002} < \frac{x-0.7515}{0.002} < \frac{0.756-0.7515}{0.002}\right)$

$= P(-1.75 < z < 2.25) = P(-1.75 < z < 0) + P(0 < z < 2.25)$

$= P(0 < z < 1.75) + P(0 < z < 2.25) = 0.4878 + 0.4599 = 0.9477$

So, the expected number of the plugs are likely to be appeared if the acceptable diameter is $0.752 \pm 0.004. = 1000\ P(0.748 < X < 0.756) = 1000 \times 0.9477 = 947.7 \approx 948.$

8. Suppose heights of student's follows normal distribution with mean 190cm and variance 80 cm. In a school of 1000 students how many would you expect to be above 200 a tall?

[Given that z= 1.118, A=0.3686].

Solution: Let Mean $\mu = 190$, Variance $v = 80$, & Standard deviation $\sigma = \sqrt{v} = \sqrt{80}$

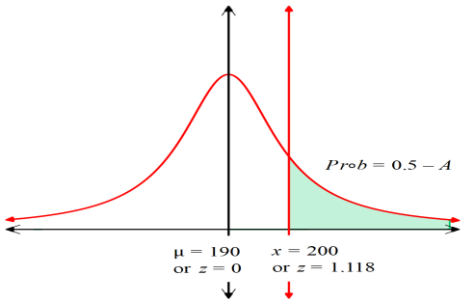
and $x = 200\ So, z = \frac{x-\mu}{\sigma} = \frac{200-190}{\sqrt{80}} = 1.118$

The probability of students expects to be above 200cm a tall is $= P(X > 200)$

$= P\left(\frac{x-190}{\sqrt{80}} > \frac{200-190}{\sqrt{80}}\right)$

$= P(z > 1.118) = 0.5 - P(0 < z < 1.118)$

$= 0.5 - 0.3686 = 0.1314$



So, the expected number of students to be above 200cm a tall $= 1000\ P(X > 200)$

$= 1000 \times 0.1314$

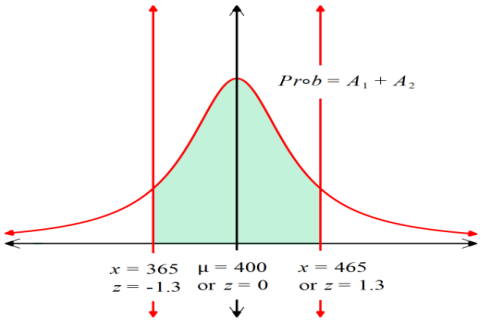
$= 131.4 \approx 131.$

9. The lifetime of an article has a normal distribution with mean 400 hours and standard deviation 50 hours. Assuming normal distribution, find the expected number of articles out of 2000 whose lifetime lies between 335 hours to 465 hours? [Given: z=1.3, A=0.4032]

Solution: Let Mean $\mu = 400$, Standard deviation $\sigma = 50$, and

$Let\ x_1 = 335\ and\ x_2 = 465$

$z_1 = \frac{x_1-\mu}{\sigma} = \frac{335-400}{50} = -1.3\ and\ \ z_2 = \frac{x_2-\mu}{\sigma} = \frac{465-400}{50} = 1.3$



hours is $= P(335 \le X \le 465)$

$= P\left(\frac{335-400}{50} \le x \le \frac{465-400}{50}\right)$

$= P(-1.3 \le z \le 1.3)$

$= 0.4032 + 0.4032 = 0.8064$

So, the expected number of articles out of 2000, whose lifetime lies between 335 hours to 465 hours$= 1{,}000P(-1.3 \le Z \le 1.3)$

$= 200 \times 0.8064$

$= 1612.8 \approx 1613.$

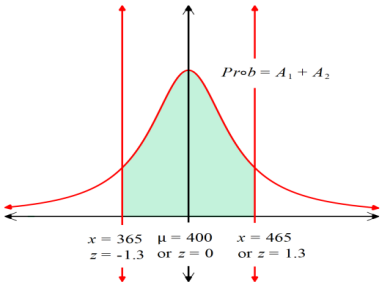
10. The lifetime of a certain component has a normal distribution with mean 400 hours and standard deviation 50 hours. Assuming normal distribution, determine approximately the

expected number of components in a sample of 1,000 components whose lifetime lies between 340 hours to 465 hours? [Given: z_1=1.2, A_1=0.3849; z_2=1.3, A_2=0.4032]

Solution: Let Mean $\mu = 400$, Standard deviation $\sigma = 50$,

and $Let\ x_1 = 340\ and\ x_2 = 465$

$z_1 = \frac{x_1-\mu}{\sigma} = -1.2\ and\ z_2 = \frac{x_2-\mu}{\sigma} = 1.3$



The probability of expected number of articles whose lifetime lies between 335 hours to 465 hours is $= P(340 \leq X \leq 465)$

$= P\left(\frac{340-400}{50} \leq x \leq \frac{465-400}{50}\right)$

$= P(-1.2 \leq z \leq 1.3) = P(-1.2 \leq z \leq 0) + P(0 \leq z \leq 1.3)$

$= P(0 \leq z \leq 1.2) + P(0 \leq z \leq 1.3)$

$= 0.3849 + 0.4032 = 0.7881$

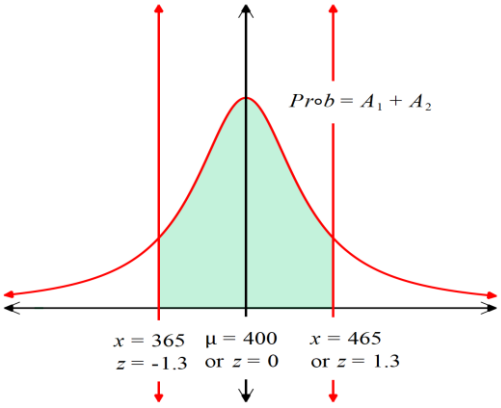
So, the expected number of articles out of 1000, whose lifetime lies between 340 hrs to 465 hours$= 1{,}000\ P(-1.2 \leq Z \leq 1.3) = 1000 \times 0.7881$

$= 788.1 \approx 788.$

11. In a sample of 1,000 cases, the mean of a certain examination is 14 and standard deviation 2.5. Assuming the distribution to be normal, find the number of students scoring between 12 and 15. [Given: z_1=0.4, A_1=0.1554; z_2=0.8, A_2=0.2881]

Solution: Let Mean $\mu = 14$, Standard deviation $\sigma = 2.5$, and $Let\ x_1 = 12\ and\ x_2 = 15$

$z_1 = \frac{x_1-\mu}{\sigma} = \frac{12-14}{2.5} = -0.8\ \ and\ z_2 = \frac{x_2-\mu}{\sigma} = \frac{15-14}{2.5} = 0.4$



The probability that a student scoring between 12 and 15 is $= P(12 \leq X \leq 15)$

$= P\left(\frac{12-14}{2.5} \leq \frac{X-14}{2.5} \leq \frac{15-14}{2.5}\right)$

$= P(-0.8 \leq Z \leq 0.4)$

$= P(-0.8 \leq Z \leq 0) + P(0 \leq Z \leq 0.4)$

$= P(0 \leq Z \leq 0.8) + P(0 \leq Z \leq 0.4)$

$= 0.2881 + 0.1554 = 0.4435$

So, the expected number of students out of 1,000 scoring marks between 12 and 15 is

$= 1{,}000\ P(12 \le X \le 15)$

$= 443.5 \approx 444.$

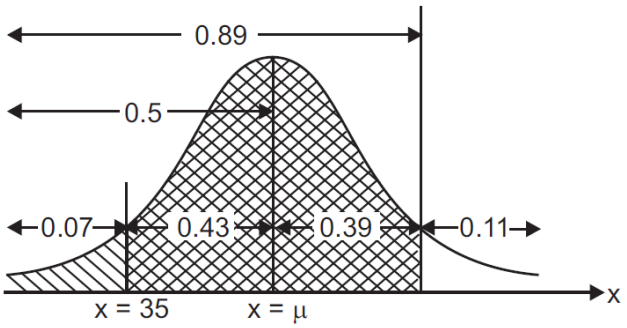
12. In a distribution, exactly normal, 7% of the items are under 35 and 89% are under 63. Find the mean and standard deviation of the distribution.

[Given that $Area\ A_1 = P(0 < z < 1.48) = 0.43$ and $Area\ A_2 = P(0 < z < 1.23) = 0.39$]

Solution: Given that $P(x < 35) = 0.07$ and $P(x < 63) = 0.89 \Rightarrow P(x > 63) = 1 - 0.89$

$P(x > 63) = 0.11.$

Let $x_1 = 35,\ x_2 = 63$ are located as shown in Fig.



When $x_1 = 35$ then $\frac{(35-\mu)}{\sigma} = -z_1\ (say)$ …(– ve sign because x=35 to the left of x = μ)

When $x_2 = 63$ then $\frac{(63-\mu)}{\sigma} = z_2\ (say)$ … (+ve sign for x=63 lies to the right of x = μ)

But given that $Area\ A_1 = P(0 < z < 1.48) = 0.43$, hence let $z_1 = 1.48$ (appx) and

$Area\ A_2 = P(0 < z < 1.23) = 0.39$, hence let $z_2 = 1.23$ (appx)

Thus, we get two simultaneous equations

$$\frac{35-\mu}{\sigma} = -1.48 \Rightarrow 35 - \mu = -1.48\,\sigma \quad \ldots i$$

$$\frac{63-\mu}{\sigma} = 1.23 \Rightarrow 63 - \mu = 1.23\,\sigma \quad \ldots ii$$

Equation (ii)-(i), we get $28 = 2.71\,\sigma \Rightarrow \sigma = 10.33\ (appx)$ and from equation (ii)

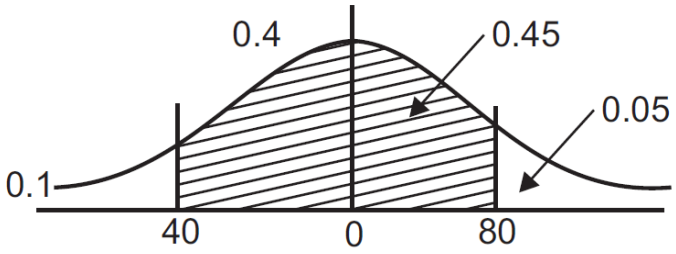
$\mu = 63 - 1.23 \times 10.33 = 50.3\ (appx).$

13. In a distribution, exactly normal, 10% of the items are under 40 and 5% are under 80. Find the mean and standard deviation of the distribution.

[Given that $Area\ A_1 = P(0 < z < 1.29) = 0.4$ and $Area\ A_2 = P(0 < z < 1.65) = 0.45$]

Solution: Given that $P(x < 40) = 10\% = 0.1$ & $P(x > 80) = 5\% = 0.05$

Let $x_1 = 40,\ x_2 = 80$ are located as shown in Fig.



When $x_1 = 40$ then $\frac{(40-\mu)}{\sigma} = -z_1\ (say)$ …(– ve sign because x=40 to the left of x = μ)

When $x_2 = 80$ then $\frac{(80-\mu)}{\sigma} = z_2\ (say)$ … (+ve sign for x=80 lies to the right of x = μ)

But given that $Area\ A_1 = P(0 < z < 1.29) = 0.40$, hence let $z_1 = 1.29$ (appx) and

$Area\ A_2 = P(0 < z < 1.65) = 0.05$, hence let $z_2 = 1.65$ (appx)

Thus, we get two simultaneous equations

$\frac{40-\mu}{\sigma} = -1.29 \Rightarrow 40 - \mu = -1.29\,\sigma \quad \ldots i$

$\frac{80-\mu}{\sigma} = 1.65 \Rightarrow 80 - \mu = 1.65\,\sigma \quad \ldots ii$

Equation (ii)-(i): we get $40 = 2.94\,\sigma \Rightarrow \sigma = 13.6\ (appx)$ and from equation (ii)

$\mu = 80 - 1.29 \times 13.6 = 57.54\ (appx)$.

14. If the r.v. X is normally distributed with mean 80 and standard deviation 5, then find

(a) $P(X > 95)$ (b) $P(X < 72)$ (c) $P(60.5 < X < 90)$

(d) $P(85 < X < 97)$ (e) $P(64 < X < 76)$

Given that $A = P(z = 3) = 0.4987, A = P(z = 1.6) = 0.4452,\ A = P(z = 3.9) = 0.5,$

$A = P(z = 2) = 0.4772,\ A = P(z = 3.4) = 0.4997, A = P(z = 1) = 0.3413$

$A = P(z = 3.2) = 0.4993, A = P(z = 0.8) = 0.2881$

Solution: Here we are given that X is normally distributed with mean 80 and standard deviation is 5. i.e. given that $\mu = 80$ and $\sigma = 5$, let $z = \frac{x-\mu}{\sigma}$

a) $P(x > 95)$: here let $x = 95$, then $z = \frac{95-80}{5} = 3$

$P(x > 95) = P(z > 3)$

$P(x > 95) = 0.5 - P(0 < z < 3)$

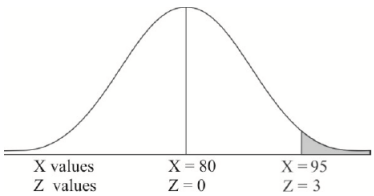
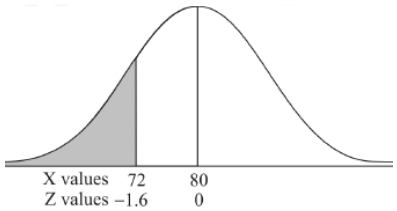


Fig. 14.6: Area to the Right of X= 95

$P(x > 95) = 0.5 - 0.4987$

$P(x > 95) = 0.0013.$

b) $P(x < 72)$: here let $x = 72$, then $z = \frac{72-80}{5} = -1.6$



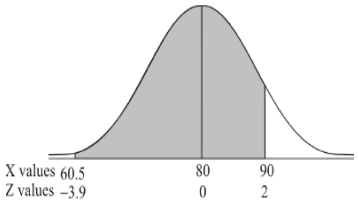
$P(x < 72) = P(z < -1.6) = P(x > 1.6)$

$P(x < 72) = 0.5 - P(0 < z < 1.6)$

$P(x < 72) = 0.5 - 0.4452$

$P(x < 72) = 0.0548.$

c) $P(60.5 < x < 90)$: let $x_1 = 60.5$, $x_2 = 90$ then $z_1 = \frac{60.5-80}{5} = -3.9$, $z_2 = \frac{90-80}{5} = 2$



$P(60.5 < x < 90) = P(-3.9 < z < 2)$

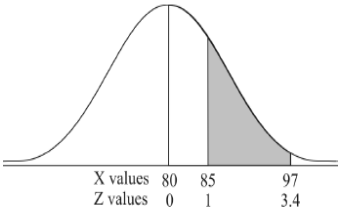
$P(60.5 < x < 90) = P(-3.9 < x < 0) + P(0 < x < 2)$

$P(60.5 < x < 90) = P(0 < x < 3.9) + P(0 < x < 2)$

$P(60.5 < x < 90) = 0.5 + 0.4772$

$P(60.5 < x < 90) = 0.9772.$

d) $P(85 < x < 97)$: let $x_1 = 85$, $x_2 = 97$ then $z_1 = \frac{85-80}{5} = 1$, $z_2 = \frac{97-80}{5} = 3.4$



$P(85 < x < 97) = P(1 < z < 3.4)$

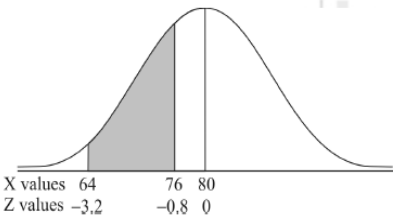
$P(85 < x < 97) = P(0 < x < 3.4) + P(0 < x < 1)$

$P(85 < x < 97) = P(0 < x < 3.4) + P(0 < x < 1)$

$P(85 < x < 97) = 0.4997 - 0.3413$

$P(85 < x < 97) = 0.1584.$

e) $P(64 < x < 76)$: let $x_1 = 64$, $x_2 = 76$ then $z_1 = \frac{64-80}{5} = -3.2$, $z_2 = \frac{76-80}{5} = -0.8$



$P(64 < x < 76) = P(-3.2 < z < -0.8)$

$P(64 < x < 76) = P(0.8 < x < 3.2)$

$P(64 < x < 76) = P(0 < x < 3.2) + P(0 < x < 0.8)$

$P(64 < x < 76) = 0.4993 - 0.2881$

$P(64 < x < 76) = 0.2112.$

CHI-SQUARE DISTRIBUTION

Definition: The quantity χ_n^2 describes the magnitude of the discrepancy between theoretical value and observed value.

If $O_1, O_2, O_3 \ldots O_k$ be the observed frequencies and $e_1, e_2, e_3 \ldots e_k$ be the expected frequencies, then the variate χ_n^2 is defined as

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i}.$$

Chi-Square Distribution is used mainly in

- Testing Hypothesis.
- Testing Independence of attributes.
- Testing the goodness of fit a model.
- The Chi-Square Distribution is denoted by χ_n^2. Here n is the parameter of the distribution also called as "*n degrees of freedom*".

Null Hypothesis: A hypothesis of "no difference" is called as null hypothesis. Null hypothesis is denoted by H_0.

For Example: $H_0 : \mu = 100$. Here the hypothesis states that there is no difference in population mean and 100. $H_0 : \mu_1 = \mu_2$ or $H_0 : \mu_1 - \mu_2 = 0$.

This hypothesis states that there is no difference between two population means. While conducting the test, some difference will be observed in sample value and hypothesized value. Whether this difference is just due to chance element, is decided in testing procedure.

Alternative Hypothesis: It is a hypothesis to be accepted in case null hypothesis is rejected. In other words, a complementary hypothesis to null hypothesis is called as alternative hypothesis. It is denoted by H_1.

For Example: If $H_0 : \mu_1 = \mu_2$ then alternative hypothesis may be $H_1 : \mu_1 \neq \mu_2$ or $H_1 : \mu_1 < \mu_2$ or $H_1 : \mu_1 > \mu_2$.

Level of Significance: Probability of rejecting null hypothesis, when it is true is called level of significance. It is denoted by α.

Examples

1. In experiment on pea breading, the following frequencies of seeds were obtained.

Round and Green	Wrinkled and Green	Round and Yellow	Wrinkled and Yellow	Total
222	120	32	150	524

Theory predicts that the frequencies should be in proportion 8:2:2:1. Examine the correspondence between theory and experiment. [Given: at $\alpha = 5\%$ level of significance expected/table value is $\chi_3^2 = 7.815$, (d.o.f.=3)]

Solution: H_0: There is a good (strong) correspondence between theory and experiment.

H_1: There is a very low correspondence between theory and experiment.

From the given data expected frequencies are (8+2+2+1=13)

Observed frequencies	222	120	32	150
Expected frequencies	$\frac{8*524}{13} = 323$	$\frac{2*524}{13} = 81$	$\frac{2*524}{13} = 81$	$\frac{1*524}{13} = 40$

O_i	e_i	$(O_i - e_i)^2$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
222	323	-101	10201	31.582
120	81	39	1521	18.778
32	81	-49	2401	29.642
150	40	110	12100	302.5
				382.5

Here $k = 4$, *Degree of freedom is* $k - 1 = 4 - 1 = 3$.

$$\chi_{k-1}^2 = \chi_3^2 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 382.5.$$

Since the calculated value (382.5) is greater than the table value (7.815) for 3 degrees of freedom at a 5% significance level.

Reject H_0.

Conclusion*: There is a very low correspondence between theory and experiment.*

2. Among 64 offspring of a certain cross between guinea pigs 34 were red, 10were black and 20 were white. According to genetic model, these numbers should be in the ratio 9:3:4. Are the

data consistent with 5% level? [Given: at $\alpha = 5\%$ level of significance expected/table value is $\chi_2^2 = 5.991$, (d.o.f.=2)]

Solution: H_0: The data is consistent.

H_1: The data is not consistent

From the given data expected frequencies are (9+3+4=16)

Observed frequencies	34	10	20
Expected frequencies	$\frac{9}{16} \times 64 = 36$	$\frac{3}{16} \times 64 = 12$	$\frac{4}{16} \times 64 = 16$

O_i	e_i	$(O_i - e_i)^2$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
34	36	-2	4	0.111111
10	12	-2	4	0.333333
20	16	4	16	1
				1.444444

Here $k = 3$, *Degree of freedom is* $k - 1 = 2$

$$\chi_{k-1}^2 = \chi_2^2 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 1.444.$$

Since the calculated value (1.444) is greater than the table value (5.991) for 2 degrees of freedom at a 5% significance level.

Accept H_0.

Conclusion*: The data is consistent.*

3. A sample analysis of examination results of 500 students was made. The observed frequencies are 220, 170, 90, 20 and the numbers are in the ratio 4:3:2:1 Find χ^2.

Solution: From the given data expected frequencies are (4+3+2+1=10)

Observed frequencies	220	170	90	20
Expected frequencies	$\frac{4}{10}500 = 200$	$\frac{3}{10}500 = 150$	$\frac{2}{10}500 = 100$	$\frac{1}{10}500 = 50$

O_i	e_i	$(O_i - e_i)^2$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
220	200	20	400	2
170	150	20	400	2.666667
90	100	-10	100	1
20	50	-30	900	18
				23.66667

Here $k = 4$, Degree of freedom is $k - 1 = 3$

$$\chi^2_{k-1} = \chi^2_3 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 23.67.$$

4. A set of five similar coins is tossed 210 times and the result is

No. of Heads	0	1	2	3	4	5
Frequency	2	5	20	60	100	23

Test the hypothesis that the data follow a binomial distribution. [Given: at $\alpha = 5\%$ level of significance expected/table value is $\chi^2_5 = 11.07$, (d.o.f.=5)]

Solution: H_0: The data follows a binomial distribution.

H_1: The data does not follow a binomial distribution.

Let probability of getting a Head is $p = \frac{1}{2}$ and Tail is $q = \frac{1}{2}$, also let $N = 210$ and $n = 5$.

The probability of getting x heads is given by the binomial formula

$$P(x) = nC_x\, p^x q^{n-x} = \frac{n!}{(n-x)! \cdot x!}\, p^x q^{n-x}$$

$$P(0) = 5C_0\, p^0 q^{5-0} = \frac{5!}{(5-0)!\cdot 0!}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^{5-0} = \left(\frac{1}{2}\right)^5 = \frac{1}{32} = 0.03125$$

$$P(1) = 5C_1\, p^1 q^{5-1} = \frac{5!}{(5-1)!\cdot 1!}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^{5-1} = \frac{5}{32} = 0.15625$$

$$P(2) = 5C_2\, p^2 q^{5-2} = \frac{5!}{(5-2)!\cdot 2!}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{5-2} = \frac{10}{32} = 0.3125$$

$$P(3) = 5C_3\, p^3 q^{5-3} = \frac{5!}{(5-3)!\cdot 3!}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^{5-3} = \frac{10}{32} = 0.3125$$

$$P(4) = 5C_4\, p^4 q^{5-4} = \frac{5!}{(5-4)!\cdot 4!}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^{5-4} = \frac{5}{32} = 0.15625$$

$P(5) = 5C_5\, p^5 q^{5-5} = \frac{5!}{(5-5)!\cdot 5!}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^{5-5} = \frac{1}{32} = 0.03125$

O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
2	$210 * 0.03125 = 6.56$	20.7936	3.169756
5	$210 * 0.15625 = 32.81$	773.3961	23.57196
20	$210 * 0.3125 = 65.63$	2082.097	31.72477
60	$210 * 0.3125 = 65.63$	31.6969	0.482964
100	$210 * 0.15625 = 32.81$	4514.496	137.5951
23	$210 * 0.03125 = 6.56$	270.2736	41.20024
			237.7448

Here $k = 6$, Degree of freedom is $k - 1 = 5$

$$\chi^2_{k-1} = \chi^2_5 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 237.744.$$

Since the calculated value (237.74) is greater than the table value (11.07) for 5 degrees of freedom at a 5% significance level.

Reject H_0.

***Conclusion**: The data does not follow a binomial distribution.*

5. A die is tossed 60 times and frequency of each face is included below

Face x_i	1	2	3	4	5	6
Frequency	5	7	5	14	13	16

Assume that the die is fair and apply Chi-square test of goodness of fit at 0.05 level of significance. [Given: at $\alpha = 5\%$ level of significance expected/table value is $\chi^2_5 = 11.07$, (d.o.f.=5)]

Solution: H_0: The die is fair.

H_1: The die is not fair.

If the die is fair then probability of each outcome is same and it is $p_i = \frac{1}{6}$, let $N = 60$ and

$\mathrm{E(x)} = e_i = \text{Expected frequency} = N\,P(x_i) = 60 * \frac{1}{6} = 10$

x_i	O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
1	5	10	25	2.5
2	7	10	9	0.9
3	5	10	25	2.5
4	14	10	16	1.6
5	13	10	9	0.9
6	16	10	36	3.6
				12

Here $k = 6$, Degree of freedom is $k - 1 = 5$

$$\chi^2_{k-1} = \chi^2_5 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 12.$$

Since the calculated value (12) is greater than the table value (11.07) for 5 degrees of freedom at a 5% significance level.

Reject H_0.

***Conclusion**: The die is not fair.*

6. A nationalized bank utilizes four windows. On a particular day 800 customers were observed inside the bank. They were given service at the different windows as following

Window number	1	2	3	4
Customers	150	250	170	230

Test whether the customers are uniformly distributed over the windows. [Given: at $\alpha = 5\%$ level of significance and expected/table value is $\chi^2_3 = 7.815$, (d.o.f.=3)]

Solution: H_0: The distribution of customers over 4 windows is uniform.

H_1: The distribution of customers over 4 windows is not uniform.

Number of windows $n = 4$ and number of customers $N = 800$, and the expected frequency per windows is $e_i = \frac{800}{4} = 200$

x_i	O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
1	150	200	2500	12.5
2	250	200	2500	12.5
3	170	200	900	4.5
4	230	200	900	4.5
				34

Here $k = 4$, Degree of freedom is $k - 1 = 3$

$$\chi^2_{k-1} = \chi^2_3 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 34.$$

Since the calculated value (34) is greater than the table value (7.815) for 3 degrees of freedom at a 5% significance level.

Reject H_0.

***Conclusion**: The distribution of customers over 4 windows is not uniform.*

7. The table below gives the number of accidents that occurred in certain factory on various day of week

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Accidents	6	4	9	7	8	10	12

Test at 5% level of significance whether accidents are uniformly distributed over a day. [Given: at $\alpha = 5\%$ level of significance and expected/table value is $\chi^2_6 = 15.592$, (d.o.f.=6)]

Solution: H_0: The accidents are uniformly distributed over a days of week.

H_1: The accidents are not uniformly distributed over a days of week

Number of days $n = 7$ & number of total accidents $N = 6 + 4 + 9 + 7 + 8 + 10 + 12 = 56$ and the expected frequency per day is $e_i = \frac{56}{7} = 8$

Day	O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
Sun	6	8	4	0.5

Mon	4	8	16	2
Tue	9	8	1	0.125
Wed	7	8	1	0.125
Thu	8	8	0	0
Fri	10	8	4	0.5
Sat	12	8	16	2
				5.25

Here $k = 7$, Degree of freedom is $k - 1 = 6$

$$\chi^2_{k-1} = \chi^2_6 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 5.25.$$

Since the calculated value (5.25) is less than the table value (15.592) for 6 degrees of freedom at a 5% significance level.

Accept H_0.

***Conclusion**: The accidents are uniformly distributed over a days of week.*

8. The demand for a particular spare part in a factory was found to vary from day to day. In a sample study. The following information was obtained

Days	Sun	Mon	Tue	Wed	Thu	Fri
No. of parts demanded	1124	1125	1110	1120	1126	1115

Test the hypothesis that the number of parts demanded does not depend on the day of the week.

[Given: at $\alpha = 5\%$ level of significance and expected/table value is $\chi^2_5 = 11.07$, (d.o.f.=5)]

Solution: H_0: The number of parts demanded does not depend on the day of the week.

H_1: The number of parts demanded depend on the day of the week.

Total number of parts demanded during a week ($n = 6$ days) is $N = 6720$.

Expected frequencies: $e_i = \frac{Total\ sum}{6} = \frac{6720}{6} = 1120$

Day	O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
Sun	1124	1120	16	0.0142

Mon	1125	1120	25	0.0223
Tue	1110	1120	100	0.0892
Wed	1120	1120	0	0
Thu	1126	1120	36	0.0321
Fri	1115	1120	25	0.0223
				0.1801

Here $k = 6$, Degree of freedom is $k - 1 = 5$

$$\chi^2_{k-1} = \chi^2_5 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 5.25.$$

Since the calculated value (0.1801) is less than the table value (11.07) for 6 degrees of freedom at a 5% significance level.

Accept H_0.

***Conclusion**: The number of parts demanded does not depend on the day of the week.*

9. Fit a Poisson distribution to the following data and test the goodness of fit by applying chi-square test.

x	0	1	2	3	4
f	155	157	58	22	8

[Given: at $\alpha = 5\%$ level of significance and expected/table value is $\chi^2_4 = 9.488$, (d.o.f.=4)]

Solution: H_0: Given data is good for Poisson distribution.

H_1: Given data is not good for Poisson distribution

By using Poisson distribution $P(r) = \frac{e^{-z}z^r}{r!}$ Where z is Mean/Average.

Total sum: $N = 155 + 157 + 58 + 22 + 8 = 400$

$$Mean: z = \frac{\sum fx}{\sum f} = \frac{(0\times155)+(1\times157)+(2\times58)+(3\times22)+(4\times8)}{155+157+58+22+8} = 0.9275$$

Expected frequencies: $e_i = N \times P(x_i)$

$$P(0) = \frac{e^{-z}z^0}{0!} = e^{-0.9275} = 0.3955$$

$$P(1) = \frac{e^{-z}z^1}{1!} = 0.9275\ e^{-0.9275} = 0.3668$$

$P(2) = \frac{e^{-z} z^2}{2!} = e^{-0.9275} = 0.1701$

$P(3) = \frac{e^{-z} z^3}{3!} = e^{-0.9275} = 0.0525$

$P(4) = \frac{e^{-z} z^4}{4!} = e^{-0.9275} = 0.0121$

x	O_i	$e_i = N\,P(x_i)$	$(O_i - e_i)^2$	$\frac{(O_i - e_i)^2}{e_i}$
0	155	158	9	0.05696
1	157	148	81	0.5472
2	58	68	100	0.4705
3	22	21	1	0.0475
4	8	5	9	1.8
				3.922

Here $k = 5$, Degree of freedom is $k - 1 = 4$

$$\chi^2_{k-1} = \chi^2_4 = \sum_{i=1}^{k} \frac{(O_i - e_i)^2}{e_i} = 3.922.$$

Since the calculated value (3.922) is less than the table value (9.488) for 4 degrees of freedom at a 5% significance level.

Accept H_0.

***Conclusion**: Given data is good for Poisson distribution.*

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